I-MAP: LOGBOOK (AI USAGE)

TSMK

Universiteit van Antwerpen
Toegepaste Ingenieurswetenschappen (Elektronica-ICT)

LLM request

⚠ ChatGPT was used for this prompt.

I have a car module that i want to make. It uses a

- Gens ace 4S Lipo Battery 5000mAh 14.8V 50C 4S Bashing RC Battery as its power source
- Corally Kuron 825 4-Pole 2050KV Brushless Sensorless Electric Motor (and its driver) to drive the wheels
- A Teensy 4.1
- A NVIDIA JETSON
- Many 3V3 and 5V pheripherals

My calculations state that (except for the motor) everything would use around 2.7A of current. To be sure that it isn't exceeded I plan to provide a 4A 5V source to these modules (except for the motor) using the TMDC 20-2411.

This would allow me to seperate the battery and motor on a seperate PCB.

Does this provide any benifit (regarding to noise on the ground plane for example)?

LLM resopnse

Short answer: **yes, it can provide real benefits**, but *only if it's done thoughtfully*.
Simply splitting things onto another PCB doesn't magically fix noise—but **power-domain separation + proper grounding absolutely helps** in systems like yours.
Let's break it down in a practical, no-hand-waving way.

1. Your system is *exactly* the kind that benefits from separation

You have:

- A **high-current, high-di/dt noise monster**
  → 4S LiPo + sensorless brushless motor + ESC
- **Sensitive digital + mixed-signal electronics**
  → Jetson Nano, Teensy 4.1, 3.3 V / 5 V peripherals

That motor/ESC combo will generate:

- Large current spikes
- High-frequency switching noise
- Ground bounce
- EMI that loves to couple into logic rails

So you're right to be thinking about this early 👍

2. What separating the PCBs *actually* gives you

✅ Real benefits if done right

Separating **motor power + ESC** from **logic power** helps with:

a) Ground noise containment

- Motor currents return through **their own copper**
- Logic ground stays quieter
- Less chance of:
  - Teensy resets
  - Jetson brownouts
  - USB / UART weirdness
  - ADC noise

b) Cleaner power regulation

Using a **dedicated DC/DC (TMDC 20-2411)** means:

- Motor current spikes don't pull down your 5 V rail
- The Jetson sees a *much stiffer supply*
- You avoid "battery sag → logic reset" moments

c) EMI control

You can:

- Keep high dI/dt loops physically small
- Route motor phase currents nowhere near logic traces
- Add filtering *only where it matters*

This is huge in RC-style builds.

3. Important reality check: PCB separation alone is

not enough

Here's the part many people miss 👇

The benefit comes from power-domain separation, not PCB count.

You still must:

🔴 Use a single-point (star) ground

- Battery negative = common reference
- Motor/ESC ground and logic ground meet **at one controlled point**
- Don't daisy-chain logic ground through the motor PCB

Best practice:

```
Battery -
   |
   +-- Motor / ESC PCB
   |
   +-- DC/DC → Logic PCB
```

🔴 Avoid shared high-current return paths

- No logic signals returning through motor ground copper
- No thin traces carrying both logic return + motor noise

4. About the TMDC 20-2411 choice

This is actually a solid approach 👍

Benefits:

- Isolated or well-filtered DC/DC (depending on config)
- Handles your ~2.7 A comfortably with margin
- Gives you:
  - A clean 5 V rail
  - Room for LDOs down to 3.3 V afterward

Strong recommendation:

Add **post-regulation filtering**:

- LC or π filter after the DC/DC
- Bulk capacitance near Jetson input
- Local decoupling *per device*

Jetson Nano in particular is *very* sensitive to supply dips.

5. Motor controller (ESC) noise deserves special handling

Sensorless brushless ESCs are **notoriously noisy**.

Do this even if PCBs are separate:

- Keep motor phase wires twisted
- Keep battery → ESC loop as short as possible
- Add:
  - Low-ESR bulk caps near ESC
  - Optional TVS on battery input
- Physically distance ESC from Jetson/Teensy if possible

Mechanical separation matters just as much as electrical.

6. Summary — does it provide benefit?

✅ Yes, if:

- You separate **power domains**, not just boards
- You use star grounding
- You prevent shared return paths
- You add proper filtering

❌ No, if:

- Grounds reconnect all over the place
- Logic ground shares motor current paths
- The DC/DC output isn't filtered

TL;DR recommendation

Your plan is **architecturally sound** and aligns with how professional robotics and RC-autonomy platforms are built.

If you want, next we can:

- Sketch a **recommended grounding topology**
- Talk **Jetson Nano power best practices**
- Review **signal isolation between Teensy ↔ ESC**
- Or sanity-check your **current calculations** under load

Just tell me where you want to zoom in 🚗⚡